

Trinadh Musunuri

trinadh.musunuri@gmail.com | linkedin.com/in/trinadh-musunuri | github.com/3nadh3 | trinadh.dev

EDUCATION

Central Michigan University

Mount Pleasant, MI

Master of Science in Computer Science

Expected May 2027

- **College:** College of Science and Engineering
- **Relevant Coursework:** Advanced Algorithms, Cloud Computing, Web Technologies, Machine Learning

EXPERIENCE

IBM

Austin, TX

Software Engineer Intern, watsonx Orchestrate

May 2026 – Aug 2026

- Led design and production deployment of end-to-end Grafana observability dashboards for the AI Gateway and Channel Integrations services (Prometheus, Loki), cutting typical error-investigation time from 10–15 minutes of manual log searching to 10–20 seconds; drove migration from local dev into the SRE team’s managed repo across Dev, Staging, and Production.
- Stood up a local voice runtime and Call Detail Record (CDR) pipeline from scratch, resolving Docker/IAM credential blockers; diagnosed a correctness bug where voice traffic bypassed the AI Gateway and fixed it via structured STT/TTS logging across 3 speech-provider adapters (Deepgram, Emotech, WatsonX).
- Built and deployed 4 production AI agents on IBM watsonx Orchestrate using React and a custom Model Context Protocol (MCP) server, validated via AWS-hosted stakeholder demo.

Central Michigan University

Mount Pleasant, MI

Research Assistant

Sep 2025 – May 2026

- Replaced an evolutionary algorithm with a gradient-based optimization method, achieving **50× faster** runtime (30–60 min → under 3 min) on adversarial ML experiments.
- Conducted low-level architectural analysis on AMD Ryzen AI hardware (XDNA 2 / Krackan), applying fine-grained cycle counting and DRAM contention metrics to profile NPU execution patterns.
- Designed and executed CPU–NPU benchmarking scripts to collect high-fidelity performance data across deep learning workloads, contributing directly to hardware security research publications.
- Engineered an interactive adversarial YOLOv2 attack demo—currently in active use by students—hosted on Hugging Face Spaces with a Cloudflare Worker proxy to circumvent restrictive university firewalls.

PROJECTS

CyberGuard XAI — Phishing Detection with Explainable AI [🔗](#)

- Built a phishing email classifier using DistilBERT with LIME and SHAP explainability, exposing per-word risk-drop scores so users can interpret and challenge model decisions.
- Developed an interactive evasion guide with word swap/undo, paginated risk cards, and a live contentEditable editor showing real-time LIME score tooltips.
- Deployed React/Vite frontend to Netlify and FastAPI backend to Hugging Face Spaces and Render with a fully automated CI/CD pipeline.

SkillSwap — Full Stack Skill Matching & Messaging Platform [🔗](#)

- Built a full-stack peer skill exchange platform using the MERN stack with WebSocket-based real-time messaging and JWT-secured REST APIs.
- Deployed on cloud infrastructure with environment-secured configuration, MongoDB data models, and scalable backend architecture.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, SQL

Web & APIs: React.js, Node.js, Express.js, FastAPI, Vite, REST, WebSockets, JWT

AI & Agentic Systems: IBM watsonx Orchestrate, Model Context Protocol (MCP), PyTorch, Vertex AI, DistilBERT, YOLOv2, LIME, SHAP, Hugging Face

Observability & Infra: Grafana, Prometheus, Loki, Redis, Docker, CI/CD

Cloud: AWS (EC2, S3, IAM, RDS, Lambda), Google Cloud, IBM Cloud, Netlify, Render

Databases: MongoDB, MySQL, PostgreSQL

Tools: Git, GitHub, Postman, Bruno, OpenAPI